Resonant Inelastic X-ray Scattering by Dynamical Mean-Field Theories

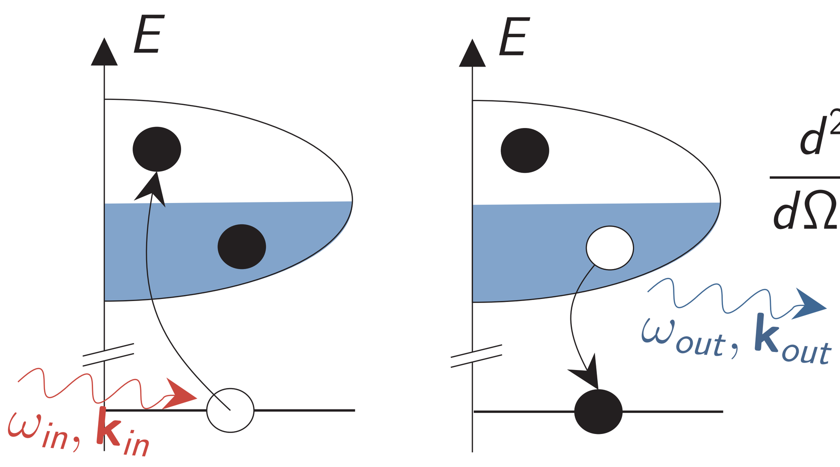
Maxime Vinteler*[1]   Coraline Letouzé[1]   Benjamin Lenz[1]   Guillaume Radtke[1]

[1]Institut de Minéralogie, de Physique des Matériaux et de Cosmochimie, Sorbonne Université, UMR CNRS 7590, Muséum National d'Histoire Naturelle, 75005 Paris, France

* Corresponding author: maxime.vinteler@sorbonne-universite.fr

Resonant Inelastic X-ray Scattering

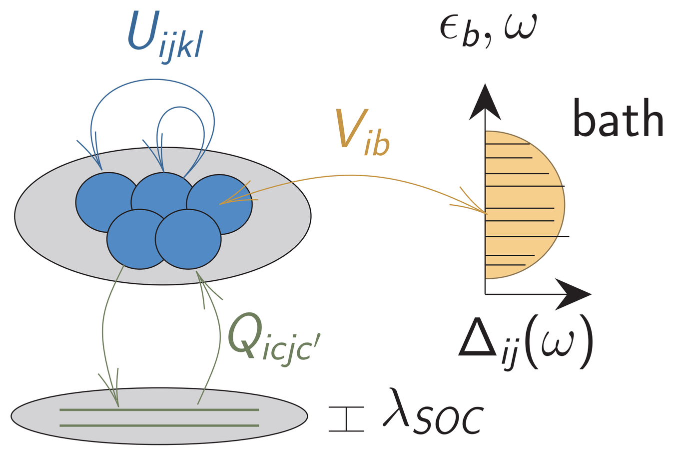
Resonant Inelastic X-ray Scattering (RIXS) is a two process spectroscopy technique with core absorption and emission. It allows to probe a lot of different excitations (phonons, magnons, *d*-*d*, charge transfer).



$$\frac{d^2\sigma}{d\Omega d\omega} \propto \sum_f \left| \sum_m \frac{\langle f | T_e^{\mathbf{q}} | m \rangle \langle m | T_a^{\mathbf{q}} | i \rangle}{\omega_{in} + E_i - E_m + i\Gamma} \right|^2 \times \delta(\omega_{in} + E_i - E_f - \omega_{out})$$

$(|i\rangle, E_i)$ is the groundstate, $(|m\rangle = T_a^{\mathbf{q}}|i\rangle, E_m)$ the intermediate absorption state and $(|f\rangle = T_e^{\mathbf{q}}|m\rangle, E_f)$ the final emission state. Γ is the quasiparticle lifetime.

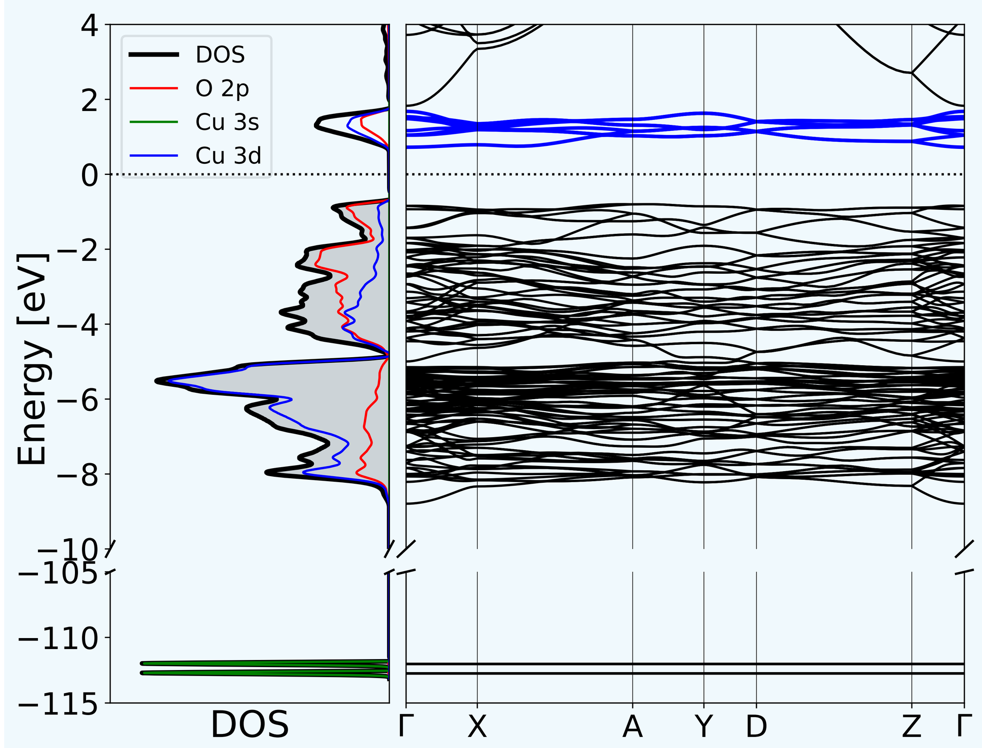
Augmented Anderson Impurity Model



$$\hat{H} = \sum_{x \in i,b,c} \epsilon_x \hat{n}_x + \sum_{i,b} V_{ib} \hat{c}_i^\dagger \hat{c}_b + h.c. + \frac{1}{2} \sum_{ijkl} U_{ijkl} \hat{c}_j^\dagger \hat{c}_i^\dagger \hat{c}_k \hat{c}_l + \frac{1}{2} \sum_{ikcc'} Q_{ickc'} \hat{c}_c^\dagger \hat{c}_i^\dagger \hat{c}_k \hat{c}_{c'} + \sum_c \lambda \hat{L}_c \cdot \hat{S}_c$$

Density Functional Theory (DFT)

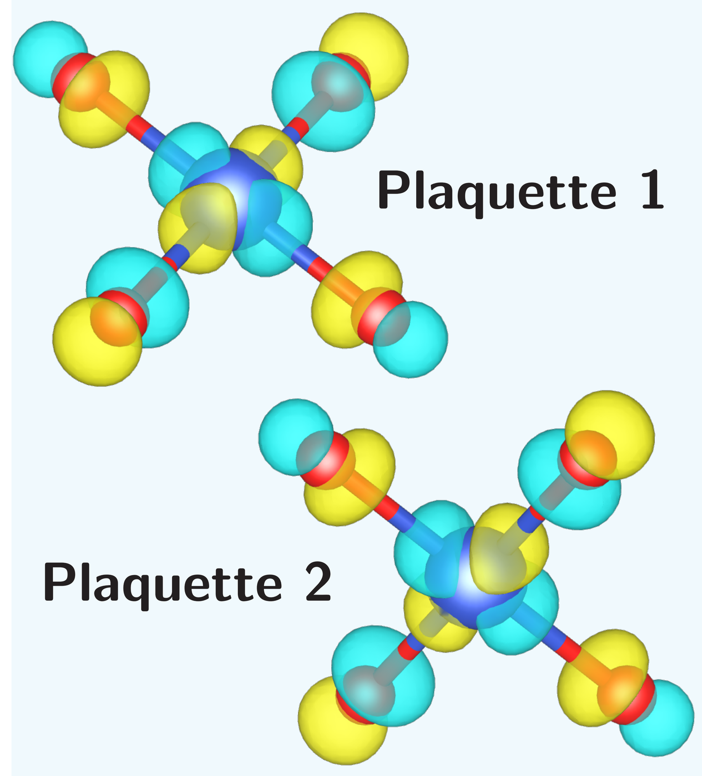
DFT+U is a mean-field, ground-state theory. It allows to compute the Bloch states (BS), the bands and the density of state (DOS) of CuO with projected Cu-3s/3d and O-2p.



- A supercell was used to model the antiferromagnetic (AF) phase. Using constrained DFT, we computed the Hubbard 3*d* interaction $U = 6.9$ eV. AF+U opens a 1.33 eV gap, in good agreement with experiment [3, 4].
- Hole states are found between 0 and 2 eV, with dominant Cu-$3d_{x^2-y^2}$ character hybridized with O-2*p* orbitals.

Wannier Functions (WF)

WF are Fourier transform of BS: $|\mathbf{R}n\rangle = \int_{BZ} d\mathbf{k}\, e^{-i\mathbf{k}\mathbf{R}} \sum_m U_{mn}^{(\mathbf{k})} |\phi_{n,\mathbf{k}}\rangle$



- We maximally localize them with respect to the $U^{(\mathbf{k})}$ gauge.
- These WFs are localized Cu-$3d_{x^2-y^2}$ orbitals centered on each Cu site **R**, hybridized with nearest O-2*p* orbitals.
- The 3*s* orbital is a core state (see DOS), we use an atomic wavefunction.

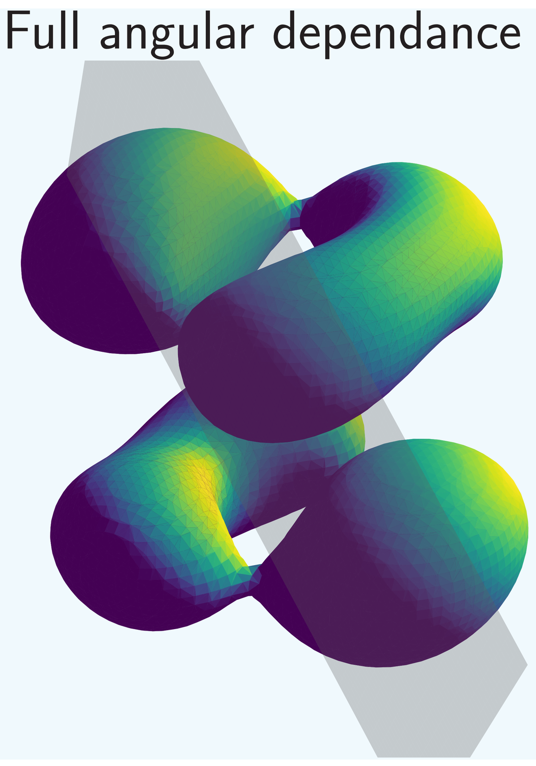
Numerical Results and Comparison with Experimental Data

- Two inequivalent copper atoms
- One $3d_{x^2-y^2}^{(i)}$ hole on each ineq. copper atom

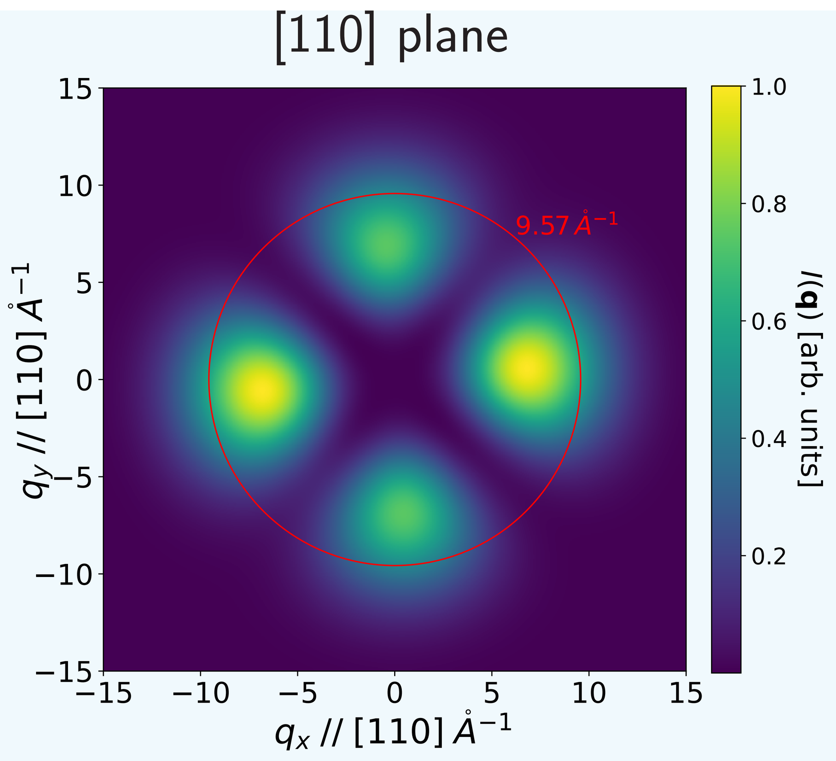
$$\longrightarrow I(\mathbf{q}) = \sum_i |\langle 3d_{x^2-y^2}^{(i)} | e^{i\mathbf{q}\mathbf{r}} | 3s^{(i)} \rangle|^2$$

- **Left:** non-orthogonal plaquettes → mixing of NRIXS contributions from inequivalent atoms.
- **Middle:** probe of the [110] plane, nearly parallel to one plaquette. Asymmetry between q_x and q_y.
- **Right:** polar plot compared with experiment (red and black dots) [2], showing good agreement. We observe a small contribution from the second plaquette.

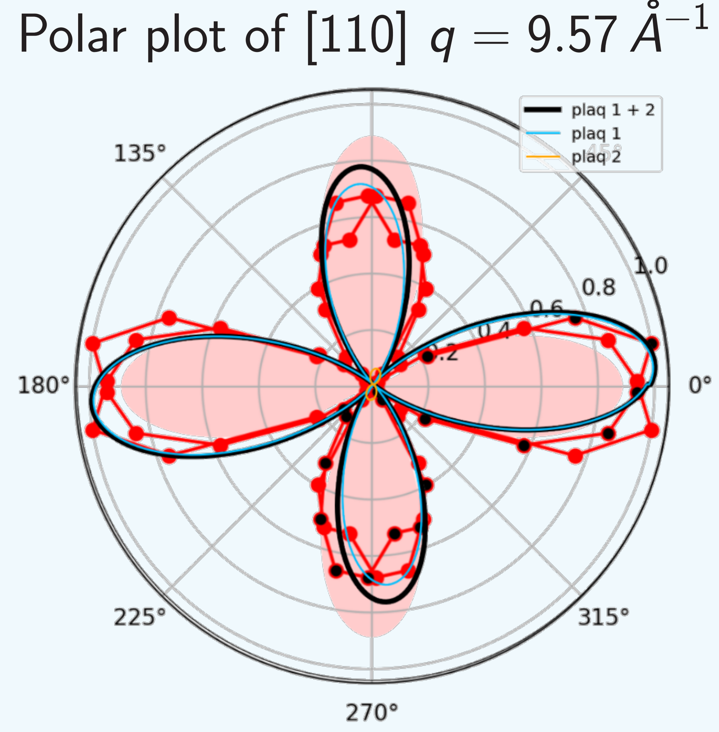
Full angular dependance



[110] plane



Polar plot of [110] $q = 9.57\ Å^{-1}$



Where does the Asymmetry come from ?

Since the second plaquette contributes only weakly to the previous figure, the asymmetry mainly originates from the fact that the probed plane is not exactly aligned with the plaquette plane. Does a plane parallel to a plaquette still exhibit such an asymmetry?

We present calculations for a plane parallel to one plaquette, with $q = 9.57\ Å^{-1}$.

- The second plaquette now contributes significantly to the overall asymmetry.
- It slightly tilts the angular distribution.
- A residual asymmetry remains in the first plaquette due to inequivalent Cu-O bonds.

Conclusion and Outlook

We developed a first-principles framework combining DFT+U and Wannier functions to compute the angular dependence of NRIXS spectra. Applied to CuO, the method reproduces the main experimental features and identifies the microscopic origin of the observed asymmetry.

Outlook:

- Future work will focus on correlated materials requiring the explicit inclusion of the density matrix in Eq. (1). We welcome suggestions of candidate systems.
- Fully self-consistent DFT+DMFT calculations also constitute also a promising extension, allowing electronic correlations to enter the Wannier functions via the feedback loop.

References

[1] H. Yavas, M. Sundermann, K. Chen, A. Amorese, A. Severing, H. Gretarsson, M. W. Haverkort, and L. H. Tjeng. *Direct imaging of orbitals in quantum materials. Nature Physics*, 15(6) :559–562, 2019.

[2] Right figure of the *Motivation* box : experimental results of orbital imaging of copper oxide at 300 K, by Victor Baledent and Jean-Pascal Rueff, with whom we are collaborating. The NRIXS spectra comes from paper [1] for NiO.

[3] D. Tahir and S. Tougaard. *Electronic and optical properties of Cu, CuO and Cu2O studied by electron spectroscopy. Journal of Physics: Condensed Matter*, 24(17):175002, apr 2012.

[4] F. Marabelli, G. B. Parravicini, and F. Salghetti-Drioli. *Optical gap of CuO. Phys. Rev. B*, 52:1433–1436, Jul 1995.

[5] We gratefully acknowledge financial support from the ANR ClusterSpecs project.